AN INTERNET AT HOME PROJECT · DATAHEARTH.NET

Hearth

A custom-built server designed to replace commonly used internet services with locally-run alternatives. Keep your data local, stay operational during disasters, and maintain full privacy control.

Proxmox | Dual Xeon | Docker Compose | 6× PCIe x16 | 3× P40 + V100

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01 Overview

🛡️ Disaster Preparedness

Functions without internet during natural or man-made disasters. Designed for prepping situations where backup power may be the only option.

⚡ Backup Power Compatible

Built to work with backup power systems, ensuring critical services remain operational when the grid goes down.

🔒 Privacy Control

Keep your data local instead of sending it to external servers. No more reliance on someone else's infrastructure.

02 Hardware

MOTHERBOARD

HUANANZHI X99 F8D Plus

PCIE SLOTS

6× PCIe x16 slots (three are x8 electrical but accept x16 cards)

CPU

Dual Xeon — Lots of cores needed for all services

GPU	QUANTITY	PURPOSE
NVIDIA P40	3	LLM inference (lacks tensor cores for image/video generation)
NVIDIA V100	1	Image and video generation (has tensor cores)

💡 **Note:** Two P40s and the V100 are optional. Removing them significantly reduces GPU costs and eliminates the need for dual power supplies and the sync adapter.

Solution

Server-class GPUs lack built-in cooling fans. The solution uses **3D-printed shrouds** with **Arctic CPU fans**. Blower fans were too loud. Single Arctic fans provided insufficient cooling, but **two Arctic fans back-to-back** increases pressure enough to cool cards quietly without sounding like a jet engine.

Configuration

An open-frame build was necessary due to:

- Two-slot-wide GPU cards cannot all install directly on the motherboard

- Cards with cooling are too large for standard cases
- eATX motherboard doesn't fit standard cases
- Dual power supply requirement

03 Stack

The main server runs **Proxmox** as the virtualization platform. Proxmox then runs several VMs, each with a **Python helper script** that lets you configure desired services and IP address, then generates a `docker-compose.yml` and installation script.

> 📝 **Proxmox Tip:** Instructions for running as open-source software and eliminating the subscription nag screen are included in `proxmox/no-nag-patch.md`.

04 Machines

🛡️ Sentinel

The Gatekeeper

Network gateway and infrastructure management

SERVICE	FUNCTION
Tailscale	Routing enabled — any connected computer can

📚 Lexicon

Media Server

Central media and content management

- **Jellyfin** — Movies & TV ✅
- **PhotoPrism** — Image management ✅
- **Kavita** — eBooks ✅
- **Audiobookshelf** — Audiobooks ✅
- **Navidrome** — Web-based music streaming ✅

SERVICE	FUNCTION
	access the Proxmox network
Backup Services	System backup management
Omada Controller	Manage Netgear routers, access points, and homelab network
VPN + Proxy	Route traffic for anonymity

- **Snapcast + Mopidy** — Smart speaker music ✅
- **Kiwix** — Offline Wikipedia ✅
- **Nextcloud** — File storage ✅
- **Gitea** — Project/code storage ✅
- **Manyfold** — STL file storage ✅
- **Headway** — Offline maps 🔧 Pending
- **Solr + Grok** — Search 🔧 Pending
- **TVHeadEnd** — DVR 🔧 Pending

💡 **Tip:** Most services support multiple instances, allowing you to separate content into **curated**, **everything**, and **adult-only** servers.

🎙️ Echo

Voice Assistant AI

AI services for Home Assistant voice control

SERVICE	FUNCTION
Ollama	Home Assistant conversation agent
Whisper	Speech-to-text conversion

Allows Home Assistant to be set up as a voice assistant without internet services.

🏠 Citadel

Home Automation

Home Assistant hub

- **Home Assistant** — Central home automation platform
- **USB Passthrough** — Zigbee/Z-Wave adapter for light control

🎯 Goal: Migrate away from IoT devices that require internet

Slower than Amazon/Google/OpenAI but keeps everything local for privacy.

🧠 Deepthought

LLM Server

Local large language model inference

SERVICE	FUNCTION
Ollama	Run LLM models locally
Open WebUI	Web interface for model interaction

Use Cases: Run RooCode or other AI agents without internet • Uncensored and unmonitored usage

🎨 Canvas

Creative AI

Image and video generation

SERVICE	FUNCTION
ComfyUI	AI image/video generation interface

Benefits: Uncensored and unmonitored creative generation • Full local control over outputs

05 Structure

```
DataHearth/
├── Citadel/          # Home Assistant configuration
├── deepthought/      # LLM server setup
├── echo/             # Voice assistant AI
├── lexicon/          # Media server configuration
├── proxmox/          # Hypervisor documentation
├── Sentinel/         # Gateway services
```

```
└── readme.md         # Documentation
```